

(Please write your Enrollment Number)

Enrollment No. _____

MID-TERM EXAMINATION

(Course Name: B. Tech. (ECE, ECE AI-1 & ECE AI-2) (Semester: 1st)
(October 2024) OFF LINE mode

Subject Code: BEC 103	Subject: ELECTRONICS WORKSHOP
Time : 1 ½ Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
a)	Explain few active and passive components you are familiar with. Why they are called passive and active components.		UNIT 1 & 2
b)	How Zener diode works? What is special about this diode? Can you use it as voltage regulator? Explain.		
c)	Explain the operation of bipolar junction transistor qualitatively. Can you say they are two diodes back to back? Explain.		
d)	What is clamper circuit? What components are involved in this? Explain the circuit functionality.		
UNIT I			CO Mapping
Q2	Attempt any two parts	(5*2=10)	
a)	Explain PN junction depletion region and also diffusion current. Explain capacitances in depletion and diffusion mode.		UNIT 1
b)	Explain forward bias and reverse biased in PN junction diode. Explain break down region. Can you utilize breakdown region in any way? Explain		
c)	Explain CE configuration. Explain DC load line and Q Point.		
UNIT II			CO Mapping
Q3	Attempt any two parts	(5*2=10)	
a)	How can you use BJT as transistor? Explain with NPN and PNP transistor both.		Unit 2
b)	What is stabilization in BJT/NPN transistor? What is the need of it?		
c)	From the figure below, find I_Z and V_{out} ? 		